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	Microprocessor / Microcontroller Lab 8085/8086/8088/8031/8051/8096/8097/68000/80486 Pentium with/without PC Interface with LED/LCD Display, Interfacing & Study Cards.
	Embedded System Lab 8051/89c51 Embedded Trainer, PIC Microcontroller
	Digital / Analog / IC / Network Lab Digital & Analog Bread Board Trainers, IC Trainers, Gates, Flip Flops, Shift Registers, Counters, Oscillators, Multivibrators, Op-Amps, Transistors, Filters, Amplifiers, Bridges, Network Theorems, Rectifier, Linear, Discrete Component Trainer etc.
	Control Systems Lab PID Controller, Synchro Transmitter & Receiver, AC/DC Motor Position & Speed Controller, AC/DC Servo Motor Controller, Stepper Motor Controller, Temperature Systems Simulator, PED, Magnetic Amplifier, Light Intensity Controller, PLC Trainer, Relay Controller, Universal AC/DC Motor Speed Controller, Stroboscope etc.
	PCB Design Lab (Single / Double Sided) Drilling Mic, Etching Mic, Curing Mic, Roller Tinning Mic, Shering Mic, U.V. Exposure Mic, Film Making Mic, Photoresist Dip Coating Mic, Dry Film Laminator Mic, Dye/Developer Mic, PTH Plant, Transformer Winding Mic, Art Work Table, Reprographics Camera Chemicals & PCB Design Software.

INSTRUMENTATION LAB

COMMUNICATION LAB

CONSUMER ELECTRONICS LAB

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 DEALERS ENQUIRY SOLICITED

log-in access control to servers.

The most important measure to secure servers is at the physical level, with access allowed only to authorised people. All servers are housed in the two server rooms, which normally remain locked and the keys are only with the senior systems administrator during each of the two shifts.

For login-based access control, only senior system administrators have passwords for root access and junior administrators have limited access (using sudo) to servers. All systems administrators have to login with their own user names and associated passwords. Normal users have no administrative access to any of the servers.

All users, including administrators, are authenticated through LDAP servers for network access. Administrators are assigned different privileges on the server by the senior system administrator. All cannot work on all servers. Faculty members and staff have access only to data in their home directories, Prowess database and common shared data in the data exchange server. The faculty members and staff can view data on the intranet server.

All the users in both the networks (staff, faculty and students) are authorised with different levels of authorisation. The LTSP users (staff and faculty) are authenticated centrally through the LDAP server. All students are centrally authenticated through the RADIUS server. Also, for wireless access, MAC address bindings are used to assign the IP address to the authorised students only.

Security from intruders

The gateway and proxy servers that connect to the Internet run the Shoreline firewall (commonly known as Shorewall). All ports on these two machines are closed on the public interface. Even local internal servers normally have only need-based ports open. Besides all the above restrictions, the ERP, Web and mail servers allow SSH access only from one IP address.

Training day

Setting up this kind of infrastructure that uses a lot of free software solutions, needed a lot of expertise. The institute initially hired a consulting company. The primary role of the company was to set up the LTSP server and deploy around 20 thin clients.

This deployment also meant a learning process. The staff members needed to be trained in using this new technology. The secondary responsibility of the company was to train the system administrators who later became trainers themselves. It was now mandatory for all system administrators to either be certified before joining or obtain Red Hat Certification within one year.

All training for staff and faculty members was done in-house by system administrators. The institute is now planning to use its experience to train outsiders so that others can also reap the benefits of Linux and Free Software technologies. ■